Resolving Token-Space Gradient Conflicts: Token Space Manipulation for Transformer-Based Multi-Task Learning

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Background

- **Multi-task learning (MTL)** promotes shared representation learning across tasks to improve generalization and efficiency, yet it often suffers from negative transfer.
- It is commonly known that negative transfer is caused by conflicting objectives between tasks.

Definition 1 (Conflicting gradients)**.** *Define g_i as the gradient of task τ_i with respect to the shared parameters Θ_s as $g_i = \nabla_{\Theta_s}\mathcal{L}_i(\Theta_s, \Theta_i)$. Let g_i and g_j represent the gradients for a pair of tasks τ_i and τ_j where $i \neq j$. If $g_i \cdot g_j \leq 0$, these two gradients are termed conflicting gradients.*

- However, recent studies show that negative transfer does not necessarily arise from gradient conflicts in parameter space.
- Some methods, such as RECON, address gradient interference by duplicating entire Transformer blocks per task, which inflates parameter counts and GPU memory in large models.

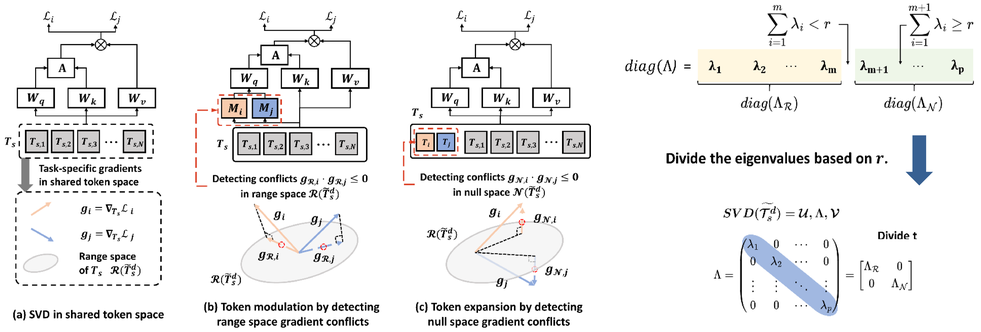
Motivation

- How about we address negative transfer directly in **token space**? This gives two advantages:

1. Finer control than parameter updates with adaptability to both range-space and null-space conflicts.
2. A fixed backbone with lightweight token-space manipulation, yielding consistent multi-task gains with minimal overhead.

Main Method

- **Token-Space Gradient Conflict Measurement.** Unlike prior methods that detect conflicts at the parameter level, this work introduces a novel approach that measures gradient conflicts directly in the token space, allowing finer-grained analysis.
- **Row Space vs. Null Space Conflict Decomposition.** It decomposes gradient conflicts into two types: conflicts within the token encoding subspace (row space), and conflicts orthogonal to it (null space), reflecting inconsistencies that lie outside the representational capacity of the network.



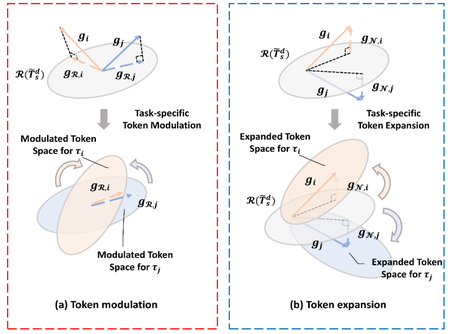
(a) SVD in shared token space (b) Token modulation by detecting range space gradient conflicts (c) Token expansion by detecting null space gradient conflicts

- **Adaptive Token Manipulation Based on Conflict Location.** For row space conflicts, where the encoder has sufficient capacity but directions misalign, the model applies Token Modulation to adjust token representations. For null space conflicts, which indicate missing capacity, it uses Token Expansion to enrich the feature space with task-relevant components.

Algorithm 1: DTME-MTL

Data: Task $\{\tau_i\}_{i=1}^K$, Loss function $\{\mathcal{L}_i\}_{i=1}^K$, Dataset $\{\mathcal{X}_i, \mathcal{Y}_i\}_{i=1}^K$, Shared tokens $T_s^{1:d} = \{T_s^{1:d}\}_{i=1}^N$, Depth of the Network D

for *each layer of the network* $(d \leftarrow 1$ *to* $D)$ **do**
- Get tokens $\{T_s^{1:d}\}_{i=1}^n$ for the layer d corresponding to input $\{\mathcal{X}_i\}_{i=1}^n$
- Calculate uncentered covariance. $\widetilde{T}_s^d = \frac{1}{n}\sum_{i=1}^n (T_s^{1:d})(T_s^{1:d})^T$
- Singular value decomposition. $\mathcal{U}, \Lambda, \mathcal{V} = SVD(\widetilde{T}_s^d)$
- Divide range and null space. $\mathcal{U} = [\mathcal{U}_{\mathcal{R}}, \mathcal{U}_{\mathcal{N}}]$
- Projection to range space. $\{g_{\mathcal{R}i}\}_{i=1}^K = \{(\mathcal{U}_{\mathcal{R}}\mathcal{U}_{\mathcal{R}}^T)\nabla_{T_s^{1:d}}\mathcal{L}_i\}_{i=1}^K$
- Projection to null space. $\{g_{\mathcal{N}i}\}_{i=1}^K = \{(\mathcal{U}_{\mathcal{N}}\mathcal{U}_{\mathcal{N}}^T)\nabla_{T_s^{1:d}}\mathcal{L}_i\}_{i=1}^K$
- **if** $g_{\mathcal{R}i} \cdot g_{\mathcal{R}j} \leq 0$ **then**
  - Insert token modulators $\mathcal{M}_i$ and $\mathcal{M}_j$
- **if** $g_{\mathcal{N}i} \cdot g_{\mathcal{N}j} \leq 0$ **then**
  - Insert task-specific tokens $\mathcal{T}_i$ and $\mathcal{T}_j$



(a) Token modulation (b) Token expansion

Experimental Results

- **Consistent gains on two benchmarks**. TM+TE lifts multi-task scores over the plain multi-task baseline.
- **Tiny parameter footprint.** All improvements come with ≤ 0.3 % extra parameters, keeping the model practically the same size as the shared-encoder baseline.
- **Complementary benefits.** TM tackles range-space conflicts, TE handles null-space conflicts, and combining both outperforms using either one alone across every task evaluated.

Impact of Token Modulation (TM) and Token Expansion (TE)

Model	NYUD-v2				PASCAL-Context				
	Semseg mIoU ↑	Depth RMSE ↓	Normal mErr ↓	Edge odsF ↑	Semseg mIoU ↑	Parsing mIoU ↑	Saliency maxF ↑	Normal mErr ↓	Edge odsF ↑
Baseline (ST)	39.35	0.6611	22.14	59.68	67.96	58.90	83.76	15.65	47.70
Baseline (MT)	34.13	0.6732	22.51	55.30	54.47	51.48	82.04	16.22	41.28
TM	37.85	0.6490	21.75	56.92	64.28	55.10	83.02	15.40	45.80
TE	37.25	0.6553	21.87	57.00	60.51	54.00	82.85	15.55	44.98
TM+TE	38.27	0.6370	21.64	57.90	66.18	56.29	83.41	15.26	47.00
Gain (vs. MT)	△ **4.14**	△ **0.0362**	△ **0.87**	△ **2.60**	△ **11.71**	△ **4.81**	△ **1.37**	△ **0.96**	△ **5.72**
△ m ↑	**0.044**				**-1.289**				
#Param ↑ (%)	**0.24**				**0.30**				

- Applying TM + TE where tasks conflict the least yields minimal additional benefit, confirming that expansion is most effective precisely in layers with the strongest conflicts.

Model	NYUD-v2				PASCAL-Context				
	Semseg mIoU ↑	Depth RMSE ↓	Normal mErr ↓	Edge odsF ↑	Semseg mIoU ↑	Parsing mIoU ↑	Saliency maxF ↑	Normal mErr ↓	Edge odsF ↑
TM+TE	**38.27**	**0.6370**	**21.64**	**57.90**	**66.18**	**56.29**	**83.21**	**15.26**	**47.00**
TM+TE (Random)	36.88	0.6567	22.27	56.30	62.12	54.43	82.95	15.55	45.80
TM+TE (Reverse)	34.71	0.6898	22.59	55.80	57.84	52.82	82.75	15.74	43.20

- DTME-MTL tops Table on Taskonomy (11 tasks, ViT-B) it cuts negative transfer, boosting average task accuracy by +4.67% with only +0.118 % extra parameters—best overall multi-task score among all optimization baselines.

Comparison with previous multi-task optimization.

Task Metric	DE L1 Dist. ↓	DZ L1 Dist. ↓	EO L1 Dist. ↓	ET L1 Dist. ↓	Key2D L1 Dist. ↓	Key3D L1 Dist. ↓	N L1 Dist.	PC RMSE ↓	R L1 Dist. ↓	S2D L1 Dist. ↓	S25D L1 Dist. ↓	$\triangle_m$ ↑ (%)
ST	0.0199	0.0195	0.1085	0.1714	0.1633	0.0872	0.2715	0.7586	0.1503	0.1742	0.1504	0.00
GD	0.0187	0.0188	0.1301	0.1757	0.1733	0.0942	0.3076	0.7991	0.1826	0.1902	0.1652	- 7.83
GradDrop [8]	0.0315	0.0242	0.1390	0.1776	0.1778	0.0976	0.4564	0.8644	0.2088	0.1995	0.1752	- 26.11
MGDA [47]	-	-	-	-	-	-	-	-	-	-	-	-
UW [31]	0.0190	0.0190	0.1308	0.1758	0.1734	0.0945	0.3109	0.8009	0.1840	0.1906	0.1657	- 8.43
DWA [36]	0.0186	0.0187	0.1294	0.1759	0.1735	0.0938	0.2788	0.7943	0.1805	0.1902	0.1640	- 6.45
PCGrad [66]	0.0217	0.0192	0.1298	0.1775	0.1714	0.0939	0.2856	0.7985	0.1817	0.1927	0.1595	- 8.29
CAGrad [33]	0.0219	0.0203	0.1314	0.1800	0.1665	0.0932	0.3039	0.8121	0.1874	0.1953	0.1673	- 10.57
IMTL [35]	0.0210	0.0192	0.1282	0.1772	0.1719	0.0936	0.2468	0.7784	0.1734	0.1943	0.1647	- 6.17
Align-MTL [48]	0.0189	0.0193	0.1254	**0.1728**	**0.1664**	0.0914	0.3524	0.8640	0.1938	0.1889	0.1582	- 9.41
Nash-MTL [43]	0.0201	0.0184	0.1248	0.1764	0.1701	0.0921	0.2658	0.7793	0.1706	0.1914	0.1624	- 5.01
FAMO [34]	0.0188	0.0188	0.1258	0.1733	0.1642	0.0858	0.2986	0.7986	0.1826	0.1904	0.1654	- 7.87
DTME-MTL	**0.0150**	**0.0154**	**0.1193**	0.1733	0.1668	**0.0891**	**0.2038**	**0.7373**	**0.1567**	**0.1773**	**0.1517**	**+ 4.67**